

James Seidel

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EDUCATION

University of Texas at Austin, Austin, TX

May 2029

Bachelor of Science, Mechanical Engineering

Overall GPA: 4.00/4.00

Relevant Coursework: Thermodynamics, Statics, Diff. Eqns. w/ Linear Algebra, Intro to Engr. Design and Graphics

ENGINEERING EXPERIENCE

Engineering Intern | Southwest River Engineering, 06/2026 – 07/2026

- Executed hydraulic analysis using HEC-RAS to simulate flood events for 6 construction projects, verify safety of construction plans, and secure regulatory floodplain permits.
- Developed 3D surface models and plan sets using **Civil 3D**, transforming raw survey data into conceptual designs for 5+ habitat restoration projects and the redesign of 3 miles of drainage channel.
- Conducted daily site inspections and maintained technical logs to document construction progress and ensure contractor compliance with engineering specifications.

Systems Subteam Member | Texas Flight, 09/2025 – Present

- Develop a 3D-printable, missile-shaped towed sensor with a four-fin stabilization system using **SolidWorks**, designed for structural loads up to 15 lbs and 60 mph flight.
- Design an adjustable CG system for 5+ payload configurations (5–15 lb), achieving CG placement within 0.03 in. of target (0.11% of sensor length) to maintain consistent flight stability.
- Collaborate with a 20+ member multidisciplinary team in weekly meetings to develop, iteratively test, and optimize critical RC aircraft components to meet competition objectives.

Mechanical Lead | Robotathon (UT IEEE RAS), 09/2025 – 11/2025

- Led the mechanical design and fabrication of a projectile launching mechanism for a semi-autonomous robot using **Onshape** and 3D printing (awarded **Best Design** of the competition).
- Coordinated with a 10-person multidisciplinary team to integrate the launcher with autonomous maze navigation and color-sensing subsystems, communicating design needs and constraints across the team.

ENGINEERING PROJECTS

3D-Printed VTOL Drone Interceptor | Personal Project, 06/2026 – Present

- Developed a 10-part fully 3D-printable quadcopter airframe in **SolidWorks**, iterating through 10+ design revisions to optimize weight, manufacturability, and aerodynamic performance.
- Designed the propulsion system, battery configuration, structural layout, and modular component interfaces for a complete UAV platform.

Odor-Reducing Laundry Hamper | Design Lead, 10/2025 – Present

- Spearhead the structural design and manufacturing of a specialized laundry hamper in **Onshape**, engineered to actively reduce odors through airflow management and clothing storage.
- Test prototype performance by measuring impacts of airflow velocity and humidity levels on odor mitigation.

QUALIFICATIONS

- **CAD:** SolidWorks, Onshape, AutoCAD, AutoCAD Civil 3D
- **Manufacturing:** 3D Printing, Rapid Prototyping, Laser-cutting, Soldering
- **Software:** Python, Microsoft Office Suite (Excel, Word, PowerPoint)
- **Honors:** University Honors (Fall 2025, Spring 2026), National Merit Scholar Finalist (2024)